

Conjecture 00000002481: the Durfee square decomposition, its generating function and its conjugation symmetry

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2026-10-05

1 The conjecture and the reading

Statement (English, verbatim). Definition: The Durfee square: geometry of partitions. Conjecture: The geometric decomposition of a partition by its Durfee square yields a symmetric splicing: the generating function factorizes into corner pieces, and the splicing is symmetric under conjugation. (The Chinese text repeats the same key words: Durfee square, geometric decomposition of partitions, symmetric splicing.)

Reading. “Symmetric splicing” is not a standard term. We read the statement as the conjunction of three claims, which together are the classical Durfee decomposition:

- (1) *Decomposition.* If λ has Durfee square of side d , its cells split into the $d \times d$ square, a piece α to the right of the square (a partition with at most d parts) and a piece β below it (a partition with parts $\leq d$); this is a bijection $\lambda \leftrightarrow (\alpha, \beta)$ with $|\lambda| = d^2 + |\alpha| + |\beta|$.
- (2) *Factorization of the generating function into corner pieces.* $\sum_{\lambda: \text{durfee}(\lambda)=d} q^{|\lambda|} = q^{d^2} \cdot C_d(q) \cdot C'_d(q)$, where C_d, C'_d are the generating functions of the two corner pieces; $C_d = C'_d = 1/(q; q)_d$ with $(q; q)_d = \prod_{i=1}^d (1 - q^i)$; and summing over d , $\sum_{\lambda} q^{|\lambda|} = \sum_{d \geq 0} q^{d^2} / (q; q)_d^2$.
- (3) *Symmetry under conjugation.* Conjugation preserves the Durfee side and exchanges the two pieces, each conjugated; the splice of (square, α, β) conjugates to the splice of (square, β', α').

A reviewer may hold a different reading of “symmetric splicing”; we prove exactly (1)–(3). This is a known classical result, not a new one. The retrieved Wikipedia article “Durfee square” (https://en.wikipedia.org/wiki/Durfee_square) states the definition (“the largest square that is contained within a partition’s Ferrers diagram”), the generating function $P(x) = \sum_{k \geq 0} x^{k^2} / \prod_{i=1}^k (1 - x^i)^2$, where the squared factor “represents the two sections to the right and below a Durfee square of size k (being two partitions into parts of size at most k , equivalently partitions with at most k parts)”, and that “the Durfee square of a partition and its conjugate partition have the same size”. Our contribution is a complete Lean 4/Mathlib formalization.

Conventions. Partitions are Mathlib `YoungDiagrams` (Ferrers diagrams): finite sets of cells $(i, j) \in \mathbb{N} \times \mathbb{N}$ (row i , column j , both from 0) closed under moving up or left. The size $|\lambda|$ is the number of cells (`YoungDiagram.card`); the conjugate is `YoungDiagram.transpose`, $(i, j) \in \lambda' \iff (j, i) \in \lambda$; the parts are the nonzero row lengths. The empty partition is included ($d = 0$). “At most d parts” means every cell is in a row $< d$ (`RowsLt d`); we prove this is `colLen0 ≤ d`, i.e. the number of rows `rowLens.length` is $\leq d$. “Parts $\leq d$ ” means every cell is in a column $< d$ (`ColsLt d`); we prove this is `rowLen0 ≤ d`. Generating functions are formal power series in $\mathbb{Z}[[q]]$.

“ $F = 1/(q; q)_d$ ” is stated as $F \cdot (q; q)_d = 1$, which determines F because $(q; q)_d$ has constant term 1; $(q; q)_0 = 1$. The infinite sum over d is a `HasSum` in the coefficientwise (product) topology of `Mathlib (PowerSeries.WithPiTopology)`.

2 Definitions (as formalized)

Definition 1 (Durfee side). $\text{durfee}(\lambda)$ is the least i with $(i, i) \notin \lambda$ (`durfee`). We prove $(i, i) \in \lambda \iff i < \text{durfee}(\lambda)$ and the two textbook characterizations: $\text{durfee}(\lambda)$ is the largest d such that the $d \times d$ square lies in λ (`isGreatest_durfee`), and the largest s with $s = 0$ or $\lambda_s \geq s$, i.e. at least s parts are $\geq s$ (`isGreatest_durfee_rowLen`, with $\lambda_s = \text{rowLen}(s - 1)$).

Definition 2 (Cutting and splicing). $R_k(\lambda) = \{(i, j) : (i, j + k) \in \lambda\}$ removes the first k columns (`dropCols`); $L_k(\lambda) = \{(i, j) : (i + k, j) \in \lambda\}$ removes the first k rows (`dropRows`). For $r, s \in \mathbb{N}$ and diagrams α, β , the splice `glue r s α β` is

$$[0, r) \times [0, s) \cup \{(i, j + s) : (i, j) \in \alpha, i < r\} \cup \{(i + r, j) : (i, j) \in \beta, j < s\},$$

which is again a Young diagram (`glue`, membership `mem_glue`).

3 Proofs

Lemma 3 (`glue_drop`, `dropCols_glue`, `dropRows_glue`, `card_glue`). (a) If $[0, r) \times [0, s) \subseteq \lambda$ and $(r, s) \notin \lambda$, then $\lambda = \text{glue}_{r,s}(R_s \lambda, L_r \lambda)$. (b) If all cells of α are in rows $< r$ and all cells of β in columns $< s$, then $R_s(\text{glue}_{r,s}(\alpha, \beta)) = \alpha$, $L_r(\text{glue}_{r,s}(\alpha, \beta)) = \beta$ and $|\text{glue}_{r,s}(\alpha, \beta)| = rs + |\alpha| + |\beta|$.

Proof. (a) A cell $(i, j) \in \lambda$ with $i < r, j < s$ is in the rectangle; with $i < r$ and $j \geq s$ it is $(i, (j - s) + s)$ with $(i, j - s) \in R_s \lambda$; with $i \geq r$ we must have $j < s$ (otherwise $(r, s) \leq (i, j)$ gives $(r, s) \in \lambda$), so it is $((i - r) + r, j)$ with $(i - r, j) \in L_r \lambda$. Conversely each of the three pieces lies in λ . (b) The three pieces are pairwise disjoint (column $\geq s$ versus $< s$; row $\geq r$ versus $< r$), the shifts are injective, and the filters $i < r, j < s$ keep all cells under the hypotheses. $\square$

Theorem 4 ((1): `durfeeEquiv`, `durfee_decomposition`). For every d , $\lambda \mapsto (R_d \lambda, L_d \lambda)$ is a bijection from the diagrams with Durfee side d onto the pairs (α, β) with α in rows $< d$ and β in columns $< d$; its inverse is $(\alpha, \beta) \mapsto \text{glue}_{d,d}(\alpha, \beta)$. Moreover $(i, j) \in \lambda$ iff $(i < d \text{ and } j < d)$ or $(i < d, j \geq d, (i, j - d) \in \alpha)$ or $(i \geq d, j < d, (i - d, j) \in \beta)$, and $|\lambda| = d^2 + |\alpha| + |\beta|$.

Proof. If $\text{durfee}(\lambda) = d$ then $[0, d)^2 \subseteq \lambda$ and $(d, d) \notin \lambda$, so the Lemma (a) applies. A cell $(i, j) \in R_d \lambda$ has $i < d$, since otherwise $(d, d) \leq (i, j + d) \in \lambda$; similarly $L_d \lambda$ lies in columns $< d$. Conversely $\text{glue}_{d,d}(\alpha, \beta)$ contains $[0, d)^2$ and not (d, d) , so its Durfee side is d (`durfee_glue`); the Lemma (b) gives the other inverse identity and the size. $\square$

Theorem 5 ((3): `durfee_transpose`, `durfeeEquiv_transpose`, `glue_transpose`). $\text{durfee}(\lambda') = \text{durfee}(\lambda)$; $R_d(\lambda') = (L_d \lambda)'$ and $L_d(\lambda') = (R_d \lambda)'$, so the pieces of λ' are the conjugated, exchanged pieces of λ ; and $(\text{glue}_{r,s}(\alpha, \beta))' = \text{glue}_{s,r}(\beta', \alpha')$.

Proof. $(i, j) \mapsto (j, i)$ fixes the diagonal and the square, and maps the right piece to the lower piece; each identity is checked cell by cell from the membership formulas. $\square$

Theorem 6 ((2): `cnt_durfee`, `gf_durfee`, `cnt_colsLt`, `gf_rowsLt_mul_prod`, `gf_durfee_mul`, `hasSum_gf_durfee`). Let $c_P(n)$ be the number of partitions of n with property P (*cnt*; the set of Young diagrams with n cells is finite, *finite_card*) and $G_P = \sum_n c_P(n)q^n$ (*gf*). Then

$$\begin{aligned} c_{\text{durfee}=d}(n) &= \sum_{a+b=n-d^2} c_{\text{RowsLt } d}(a) c_{\text{ColsLt } d}(b) \quad (d^2 \leq n), \quad 0 \quad (d^2 > n), \\ G_{\text{durfee}=d} &= q^{d^2} G_{\text{RowsLt } d} G_{\text{ColsLt } d}, \quad G_{\text{ColsLt } d} = G_{\text{RowsLt } d}, \quad G_{\text{RowsLt } d} \cdot (q; q)_d = 1, \\ G_{\text{durfee}=d} \cdot (q; q)_d^2 &= q^{d^2}, \quad \sum_{d \geq 0} G_{\text{durfee}=d} = \sum_{d \geq 0} q^{d^2} G_{\text{RowsLt } d}^2 = \sum_{\lambda} q^{|\lambda|}. \end{aligned}$$

Proof. The counting identity is the decomposition bijection of claim (1) (`durfeeEquiv`) restricted to size n , which is a bijection onto $\bigsqcup_{a+b=n-d^2} \{\alpha \vdash a : \alpha \text{ in rows } < d\} \times \{\beta \vdash b : \beta \text{ in columns } < d\}$ by the size formula; the power series form is the coefficient formula for products and for $q^{d^2} \cdot F$. Conjugation is a size-preserving bijection between “columns $< d$ ” and “rows $< d$ ”. For the corner series $C_d = G_{\text{RowsLt } d}$: $C_0 = 1$ (only the empty diagram has no rows), and a diagram in rows $< d+1$ either lies in rows $< d$, or has exactly $d+1$ rows, in which case removing its first column (`R1`, with inverse `glued+1,1(·, ∅)`, `peel1`) is a bijection onto diagrams in rows $< d+1$ with $d+1$ fewer cells (`cnt_rowsLt_succ`). Hence $C_{d+1} = C_d + q^{d+1}C_{d+1}$, i.e. $C_{d+1}(1 - q^{d+1}) = C_d$, and induction gives $C_d(q; q)_d = 1$. Then $G_{\text{durfee}=d}(q; q)_d^2 = q^{d^2}(C_d(q; q)_d)^2 = q^{d^2}$. Finally, a partition of n has $\text{durfee}^2 \leq n$, so the q^n -coefficients of $G_{\text{durfee}=d}$ vanish for $d > n$ and sum over $d \leq n$ to $c_{\text{True}}(n)$ (fiberwise counting). $\square$

4 Lean formalization and axiom audit

File `lean/Conjecture2481/Basic.lean` (namespace `C2481`, only `import Mathlib`, 541 lines). Main declarations, with their statements as in the file:

- `durfeeEquiv d`, an equivalence $\{\mu \mid \text{durfee } \mu = d\} \simeq \{\alpha \mid \text{RowsLt } d \alpha\} \times \{\beta \mid \text{ColsLt } d \beta\}$, and `durfee_decomposition`: the cell-level splitting and $|\mu| = d^2 + |\alpha| + |\beta|$.
- `isGreatest_durfee`, `isGreatest_durfee_rowLen`, `rowsLt_iff`, `colsLt_iff`: the definitions agree with the textbook notions.
- `durfee_transpose`, `durfeeEquiv_transpose`, `glue_transpose`: (3).
- `cnt_durfee`, `gf_durfee`, `cnt_colsLt`, `gf_rowsLt_mul_prod`, `gf_durfee_mul`, `hasSum_gf_durfee`: (2).

`#print axioms` for each of these declarations (checked on the Lean v4.33.1 / Mathlib v4.33.1 toolchain of the package) reports only `[propext, Classical.choice, Quot.sound]`. There is no sorry, no `native_decide` and no added axiom.

Source

Wikipedia, “Durfee square”, https://en.wikipedia.org/wiki/Durfee_square (retrieved 2026-10-05): “An equivalent, but more visual, definition is that the Durfee square is the largest square that is contained within a partition’s Ferrers diagram.”; “It is clear from the visual definition that the Durfee square of a partition and its conjugate partition have the same size.”